

Annie Prasanna Manoharan

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Research Profile

M.S. Computer Engineering graduate with research experience using computer vision and deep learning to process and analyze annotated image and video data. Brings measurable model evaluation, reproducible preprocessing workflows, and medical-device software validation experience.

Education

University of Arkansas, Fayetteville, AR

M.S. Computer Engineering

Aug 2024 – May 2026

Coursework: Artificial Intelligence, Machine Learning, Image Processing, Data Mining

Sri Ramakrishna Engineering College, India

B.E. Electronics and Communication Engineering

Aug 2019 – May 2023

Technical Skills

Machine Learning & Vision: Python, PyTorch, deep learning, YOLO11n-Pose, OC-SORT, LSTM, Res-UNet, computer vision, image processing

Data & Research: SQL, Pandas, NumPy, data preprocessing, annotation pipelines, data validation, feature engineering, model evaluation

Software & Platforms: C++, C#, WPF/.NET, Flask, Git, Docker, LabVIEW, NI TestStand

ML / Data Tools: Scikit-learn, Apache Kafka, Apache Flink, ClickHouse, Grafana, data visualization, LangChain, FAISS

Projects

MiGa: Multi-Chicken Gait Assessment (AICV Lab Research)

YOLO11n-Pose, OC-SORT, LSTM, PyTorch

- Built a keypoint-estimation, multi-object-tracking, and many-to-one LSTM workflow to process and analyze 4,569 annotated frames and 11,422 bounding boxes of biological video data.
- Applied spatial normalization to remove translational variance across subjects; achieved 91.83% keypoint mAP, 0.93 MOTA, and 0.91 IDF1 across 55 expert-annotated sequences.

DualFit: Two-Stage Virtual Try-On (AICV Lab Research)

PyTorch, Res-UNet, flow-based warping, inpainting

- Developed a two-stage image pipeline combining flow-based garment warping with Res-UNet synthesis to improve alignment and visual realism.
- Built preprocessing and inpainting workflows, then trained and evaluated on VITON-HD: 24.9 PSNR, 0.913 SSIM, and 5.261 FID.

Real-Time Medical IoT Analytics Platform

Apache Kafka, Apache Flink, ClickHouse, Grafana, Python

- Built a sub-second-latency Flink SQL and Kafka pipeline to score high-frequency medical device data, then designed ClickHouse schemas and a Grafana visualization layer for real-time interpretability.

Professional Experience

Research Assistant, AICV Lab, University of Arkansas

Sep 2024 – Jan 2026

- Integrated YOLO11n-Pose, OC-SORT, and an LSTM network for multi-subject gait analysis from image and video data.
- Developed automated preprocessing and annotation workflows that transformed raw frames into structured model inputs and evaluation datasets.
- Trained and evaluated image-synthesis and tracking models using mAP, MOTA, IDF1, PSNR, SSIM, and FID metrics.

Data Engineering Intern, InsOps Inc.

Jul 2024 – Present

- Developed data pipelines in AWS using Glue, Lambda, EMR, and Apache Airflow for orchestration, with S3 cloud storage, for scalable ingestion, transformation, and validation of large datasets.
- Built data workflows integrating SQL Server, Azure SQL, and AWS S3, applying metadata-driven schema discovery and data-quality validation to prepare analytics-ready datasets.

Engineering Intern / Project Engineer, Soliton Technologies

Jul 2022 – Jun 2024

- Designed a C# WPF desktop application integrating C#, Python, and LabVIEW-based test workflows for medical-device verification and validation.
- Implemented reusable object-oriented components and automated testing workflows to improve development and validation efficiency.

Teaching Assistant, University of Arkansas

Sep 2024 – May 2026

- Supported 100+ students in Programming Foundations through laboratory instruction, debugging, and mentorship in C++ fundamentals.
- Supported Database Management Systems instruction for approximately 180 students, answering technical questions and assisting with assignments and assessments.